

Lab Exercise 13

Pointers

Objective

The objective of this exercise is to develop students' understanding of pointer concepts in the C programming language by enabling them to store, access, and manipulate user-input data through memory addresses. Through a simple student marks application, students will learn how pointers are used to receive input via `scanf()`, dereference memory locations to retrieve values, and perform calculations such as computing an average. This exercise also strengthens their comprehension of how variables and memory are linked in C, reinforcing the practical role of pointers in real-world programming tasks.

Video Link:

- <https://www.bing.com/videos/riverview/relatedvideo?q=Pointers+in+C+Language&mid=8C329F03C5D92343267B8C329F03C5D92343267B&FORM=VRDGAR>
- <https://www.bing.com/videos/riverview/relatedvideo?q=Pointers+Jenny+Lectures&mid=023C5C26F45A72E7126E023C5C26F45A72E7126E&FORM=VRDGAR>

Task 1

What is the output of the following code?

```
#include <stdio.h>

int main()
{
    int num = 10;
    int *ptr;

    ptr = &num;

    printf("Value of num = %d\n", num);
    printf("Address of num = %p\n", &num);
    printf("Value stored in ptr = %p\n", ptr);
    printf("Value pointed by ptr = %d\n", *ptr);

    return 0;
}
```

Task 2

Write a program that:

1. Declares an integer array

```
int arr[5] = {10, 20, 30, 40, 50};
```

2. Uses a pointer to display all elements of the array using pointer arithmetic
3. Changes the 3rd element of the array to **99** using the pointer
4. Displays the updated array.

Task 3

Write a C program that:

1. Declares an integer variable `num`
2. Declares a pointer `p`
3. Uses `scanf()` to read a number from the user **using the pointer**
4. Displays:
 - The value entered
 - The memory address where the value is stored

Task 4: Simple Student Marks Application Using Pointers

Objective

To demonstrate how pointers can be used to store, access, and modify data entered by the user.

Write a C program that:

1. Declares three integer variables to store:
 - Test 1 marks
 - Test 2 marks
 - Test 3 marks
2. Uses **pointers** to:
 - Read the marks from the user
 - Calculate the **average**
3. Displays:
 - Each mark
 - The average mark

All input and calculations must be done **using pointers**, not direct variables.